

Reflective Journal: Module 2

This lab was a good first step into actually using some of the tools that show up in machine learning. Before doing it, I had heard of things like pandas, NumPy, matplotlib, and scikit-learn, but I did not really understand how they worked together. I mostly thought machine learning was just about training a model and getting predictions. After working through this lab, I can see that a lot of the process starts before the model part even happens. You first have to load the data, organize it, look at it, graph it, and understand what the data is showing.

One thing I liked about this lab was that it used the Iris dataset, which was simple enough to follow but still showed real machine learning ideas. The dataset had 150 samples, four measurement features, and three flower species. Seeing that structure helped me understand what people mean when they talk about features and classes. The measurements like sepal length, sepal width, petal length, and petal width are the features, and the flower species are the classes. That made the terms feel a lot less abstract.

The biggest learning moment for me was probably creating the scatter plot. Looking at the table of numbers was useful, but the graph made the patterns way easier to see. Once the points were plotted by sepal length and sepal width, it became clear that setosa was easier to separate from the other two species. At the same time, versicolor and virginica overlapped more in the middle of the graph. That helped me understand why visualization matters. It is not just something extra to make the assignment look better. It actually helps you notice patterns that would be harder to catch by only reading the data.

I also started to see how each tool has its own role. Pandas made the dataset easier to work with by putting it into a DataFrame. NumPy was useful for calculations like finding the mean and standard deviation of sepal length. Matplotlib was used to create the scatter plot and bar chart. Scikit-learn provided the Iris dataset. At first, it felt like a lot of different tools at once, but once I understood the purpose of each one, the workflow made more sense.

Another thing I learned was that basic statistics are still important in machine learning. The mean sepal length gave me a quick idea of the average size in the dataset, while the standard deviation showed how spread out the values were. I used to think statistics were separate from machine learning, but now I can see that they are connected. A model is basically trying to learn from patterns in the data, so it makes sense that we should understand the numbers before jumping into modeling.

The species count bar chart was also helpful, even though it was simple. It showed that each species had 50 samples. That made me realize that checking the balance of a dataset matters. If one species had way more examples than the others, that could affect how a model learns later. Even though we did not train a model in this lab, I can already see how early data exploration can help prevent problems later on.

One challenge for me was getting used to the syntax. Some of the code was easy to understand, like using `df.head()` to see the first few rows. Other parts took more attention, especially grouping the data by species and looping through each species to plot the points. I had to remind myself that the goal is not to memorize every command right away. The goal is to understand what the code is doing and why it is useful.

Overall, this lab helped me understand that machine learning starts with understanding the data. Before this assignment, I probably would have skipped straight to wanting to build a model. Now I see why data exploration is such an important first step. If I do not understand the data, then I cannot really understand what the model is learning from it.

Going forward, I want to get better at reading graphs and asking better questions from them. For example, after seeing the overlap between *versicolor* and *virginica*, I would want to know whether petal length and petal width would separate the species better than sepal length and sepal width. I would also want to understand which features are most useful for classification.

This assignment made the machine learning process feel more realistic to me. It showed me that tools like pandas, NumPy, matplotlib, and scikit-learn are not just random libraries to memorize. They each help with a different part of working with data. The main thing I took away is that good machine learning depends on good data understanding first.